

RESEARCH INTERESTS

Asset Pricing, Financial Intermediation, Real Estate

EDUCATION

Pennsylvania State University (Smeal)
Ph.D. in Business Administration

University Park, PA
2022–2027 (expected)

University of Tokyo
M.A. in Economics
B.A. in Economics

Tokyo, Japan
2019–2020
2015–2019

PROFESSIONAL EXPERIENCE

Board of Governors of the Federal Reserve System
Dissertation Fellow, Division of Monetary Affairs

Washington, D.C.
Summer 2026

Deloitte Touche Tohmatsu LLC
Consultant, Deloitte Analytics

Tokyo, Japan
2020–2022

WORKING PAPERS

JOB MARKET PAPER

[1] **Who Holds U.S. Bonds and How It Shapes the Yield Effects of QE and QT**

How much do the bond-supply shifts of quantitative easing (QE) and tightening (QT) move Treasury yields and mortgage rates? The answer turns on the slope of aggregate bond demand: who holds bonds, how sensitive their demand is to yields, and how fast they rebalance. I estimate heterogeneous investor demand for Treasuries and the agency MBS that fund most U.S. mortgages, using sector-level portfolio holdings and transactions spanning 1985 to 2025. I measure each sector's demand against its own non-bond alternative and let its yield sensitivity and rebalancing speed vary. Most investors respond to yields weakly and slowly, so aggregate demand is inelastic. Feeding the demand system through market clearing, I find that historical QE compressed 10-year Treasury yields by 8 to 12 basis points per percent of the float removed. Because agency MBS demand prices the spread over Treasuries, the size of a program moves mortgage rates through the Treasury yield, and its composition sets the mortgage spread. Further runoff would move yields less per unit than QE did, because the sectors that absorb Treasuries now hold more of the float. An additional \$1 trillion of runoff raises 10-year Treasury yields by about 22 basis points and agency MBS yields by about 27, of which 5 is the mortgage spread.

Fellowship: *Federal Reserve Board Dissertation Fellowship 2026*
Presentations: *Federal Reserve Board 2026, Penn State 2026*

OTHER PAPERS

[1] **Rationed, Not Repriced: GSE Concessions, Mortgage Access, and House Prices**

(with Wenchuan Zhao)

Presentations: *AREUEA-ASSA 2027 (scheduled), Freddie Mac Rural Housing Symposium 2026, USI-SFI Real Estate and Urban Economics Conference 2026, Zurich-Oxford Doctoral Symposium on Real Estate Markets 2026, Econometric Society European Meeting 2026, FHFA 2026, FMA Europe 2026, UEA Summer School 2026, UEA European Meeting 2026*

- [2] **Expectations and Risk Premiums in Illiquid Real Assets** (with Jiro Yoshida)
 Presentations: *CREDA Research Symposium 2026, Quad Real Estate Conference 2026, AREUEA National 2026, CBRE 2026, MIT 2026, USC 2026, Cambridge-HKU-UF Real Estate Finance and Investment Symposium 2025*
- [3] **Climate Risk in Financial Contracts: Evidence from Leases, Loans, Bonds, and Mergers**
 Presentations: *MIT Climate and Real Estate Symposium 2024, FIRS 2024, Hitotsubashi 2023*

NOTES

- [1] **A Note on Identification of Difference-in-Discontinuities in Cross-Sectional Settings**
 Presentations: *Econometric Society Asian Meeting 2024, IAAE 2024*

POLICY PUBLICATIONS (NON-REFEREED)

- [1] Inflation and Real Estate Investment (with Jiro Yoshida) 2024
Securities Analysts Journal, 62(8), 31–39 (in Japanese)

PRESENTATIONS

REFEREED CONFERENCES

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|--|-------------------------|
| [1] AREUEA-ASSA Conference, Washington, D.C. | Jan 3, 2027 (scheduled) |
| [2] CREDA Research Symposium, Ph.D. Poster Session, Chapel Hill, NC | Oct 1, 2026 |
| [3] USI-SFI Real Estate and Urban Economics Conference, Lugano, Switzerland* | Sep 23, 2026 |
| [4] Zurich–Oxford Doctoral Symposium on Real Estate Markets, Oxford, UK* | Sep 8, 2026 |
| [5] Econometric Society European Meeting (ESEM), Dublin, Ireland* | Aug 19, 2026 |
| [6] Financial Management Association (FMA) European Conference, Braga, Portugal* | Jun 12, 2026 |
| [7] UTokyo-NUS-MIT-PSU Quad Real Estate Conference, Tokyo, Japan* | Jun 4, 2026 |
| [8] AREUEA National Conference, Washington, D.C.* | May 28, 2026 |
| [9] Urban Economics Association (UEA) Summer School, London, UK* | May 11, 2026 |
| [10] Urban Economics Association (UEA) European Meeting, Barcelona, Spain* | May 8, 2026 |
| [11] Cambridge-HKU-UF Real Estate Finance and Investment Symposium, Gainesville, FL | Oct 10, 2025 |
| [12] MIT Climate and Real Estate Symposium, Cambridge, MA | Oct 5, 2024 |
| [13] Econometric Society Asian Meeting, Hangzhou, China | Jun 29, 2024 |
| [14] International Association for Applied Econometrics (IAAE) Annual Conference, Xiamen, China | Jun 9, 2024 |
| [15] Financial Intermediation Research Society (FIRS) Conference, Ph.D. Session, Berlin, Germany | May 29, 2024 |
| [16] AREUEA-AsRES International Conference, Online | Jul 14, 2021 |

SEMINARS

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|--|-------------------------|
| [1] University of Reading* | Oct 7, 2026 (scheduled) |
| [2] Freddie Mac | Sep 24, 2026 |
| [3] University of Wisconsin–Madison* | Sep 10, 2026 |
| [4] Board of Governors of the Federal Reserve System (FRB) | Aug 10, 2026 |
| [5] Federal Housing Finance Agency (FHFA) | Jul 16, 2026 |

[6] Board of Governors of the Federal Reserve System (FRB)	Jun 15, 2026
[7] CBRE*	May 28, 2026
[8] Massachusetts Institute of Technology*	Apr 14, 2026
[9] University of Southern California*	Jan 23, 2026
[10] Hitotsubashi University	Aug 7, 2023

* Presented by co-author; internal presentations excluded.

AWARDS, GRANTS, AND SCHOLARSHIPS

AWARDS

[1] Early Graduation, University of Tokyo	2020
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GRANTS

[1] Smeal Research Grant, Pennsylvania State University (3,000 USD)	2026–2027
[2] Project Research Program Grant (IERPK2325/IERPK2413), Institute of Economic Research, Hitotsubashi University (with Keisuke Kawata) (854,000 JPY)	2023–2025
[3] Ph.D. Travel Grant, Financial Intermediation Research Society (1,000 USD)	2024

SCHOLARSHIPS

[1] Jeanne and Charles Rider Graduate Fellowship, Pennsylvania State University (2,200 USD)	2026–2027
[2] Peter E. Liberti and Judy D. Olian Scholarship, Pennsylvania State University (2,000 USD)	2025–2026
[3] World-leading Innovative Graduate Study Fellowship, University of Tokyo (840,000 JPY)	2019–2020
[4] Nikaido Scholarship, Nikaido Foundation (528,000 JPY)	2017–2019

TEACHING

INSTRUCTOR

Pennsylvania State University

[1] RM450: Contemporary Issues in Real Estate Markets (×3)	Spring 2024, Fall 2024, Fall 2025
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TEACHING ASSISTANT

Pennsylvania State University

[1] FIN455/RM475: Quantitative Analysis for Business	Spring 2025
[2] FIN460/RM460: Real Estate Financial Analysis	Fall 2023
[3] FIN470/RM470: Real Estate and Capital Markets	Fall 2023
[4] REST550: Contemporary Issues in Real Estate Markets (Graduate)	Spring 2026
[5] REST560: Real Estate Financial Analysis (Graduate)	Fall 2023
[6] REST575: Quantitative Analysis for Real Estate (Graduate)	Spring 2025
[7] RM450: Contemporary Issues in Real Estate Markets	Spring 2026

PROFESSIONAL SERVICE

AD HOC REFEREE

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|---|-----------|
| [1] Econometric Reviews | 2025 |
| [2] Journal of Real Estate Finance and Economics (×4) | 2023–2026 |

REFERENCE

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| [1] Brent W. Ambrose (Co-chair)
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